

Nowcasting Down Under: Australian GDP

Advanced Time Series, Forecasting and Nowcasting

June 2026

1 Introduction

This document presents a Dynamic Factor Model (DFM) built to nowcast Australia’s real GDP growth, adapting the Stock–Watson (1991) coincident-indicator framework covered in class to a new country and a richer indicator set. The choice of Australia was, admittedly, partly a matter of curiosity: a resource-exporting, financially open economy whose business cycle is shaped as much by commodity demand from Asia as by the domestic housing market seemed like a good test of whether the baseline model travels well outside the original US application. The model summarises the co-movement of one quarterly target series and seven monthly indicators (Table 1) into a single latent common factor f_t , estimated recursively via the Kalman filter and obtained by maximum likelihood, exactly as in `Homework2.m`. The `matrices.m` component of that baseline code was rewritten to accommodate eight observed series, a richer set of idiosyncratic dynamics, and one genuinely leading indicator, while the rest of the estimation engine was left untouched.

Table 1: Data description: variables, transformations, type and sample.

Variable	Unit	Transformation	Type	Native sample
GDP	Chained volume, 2019=100	$\Delta \log$	Hard (target)	Q3-1960 – Q1-2026
Industrial Production (IP)	Index, 2010=100	$\Delta \log$	Hard	08-1974 – 02-2026
Construction (dwellings approved)	Number, SA	$\Delta \log$	Hard	07-1986 – 04-2026
New motor vehicle sales	Units	$\Delta \log$	Hard	01-1994 – 05-2026
Employment	Thousand	$\Delta \log$	Hard	02-1978 – 02-2026
Stock index (ASX)	Index, USD	$\Delta \log$	Hard	06-2000 – 05-2026
Total Exports: Commodities	AUD billion	$\Delta \log$	Hard	01-1955 – 03-2026
Composite Leading Indicator (CLI)	Index	log	Leading ($t + 3$)	02-1966 – 04-2026

¹ ADF tests indicate that all variables are I(1) except the Composite Leading Indicator, which is I(0) by construction and is accordingly entered in log-levels rather than log-differences. Sources: GDP, IP, Employment, Construction, Vehicle sales and Exports from the ABS; the stock index from Bloomberg; the CLI from the OECD.

Although several indicators are individually available from the 1950s–1970s, the stock index (ASX) is the binding constraint, as it starts in June 2000. We include the ASX because its original daily frequency provides higher-frequency information that may help capture more timely changes in economic activity. To incorporate it consistently into the monthly dynamic factor model, the daily series was converted into a monthly series following the aggregation guidelines discussed in class. The model is therefore estimated on the monthly sample from **June 2000 to May 2026**, with quarterly GDP and missing observations handled by the Kalman filter.

2 Indicator Selection

The selection was guided more by heuristic, country-specific reasoning than by a formal variable-selection procedure – in the spirit of what was asked of us. Industrial production and employment are the standard, textbook contemporaneous proxies for aggregate activity, and were kept essentially for comparability with the baseline specification. Beyond these, however, we tried to let the choice of indicators speak to what actually drives the Australian cycle. Dwelling approvals were included because Australia’s business cycle is, to an unusual degree, a housing cycle: residential construction has historically been one of the most cyclically sensitive components of Australian GDP, and approvals lead actual building activity by several months. New motor vehicle sales were added as a proxy for private consumption of durables, which tends to be one of the first things households cut back on – or splurge on – as sentiment shifts. Given that Australia is a heavily resource-export-dependent economy (iron ore, coal and LNG account for a large share of export revenue), total commodity exports were a natural inclusion to capture the external, terms-of-trade-driven side of the cycle that a purely domestic indicator set would miss. The ASX stock index was added to reflect the financial and corporate sector’s own read on prospects. Finally, the OECD Composite Leading Indicator was included not as a contemporaneous variable but explicitly as a *leading* one, entering the model with a three-month lead on the factor (f_{t+3}) rather than loading on f_t like the rest – this is our route to the model’s leading-indicator extension.

2.1 Validation: Principal Component Analysis

Having motivated our indicator choices on heuristic grounds, we tested these intuitive arguments by using PCA to check whether the data supports our story. Figure 1 finds the linear combination of the indicators that explains the largest share of their joint variables, i.e. the first principal component f_t under the assumption that each series is a noisy, partial signal of a latent business-cycle state. This makes the resulting loadings a diagnostic for variable selection, as the series whose movements are mostly idiosyncratic and unrelated to the broader cycle will load weakly on f_t , while a series that tracks this cycle will load strongly.

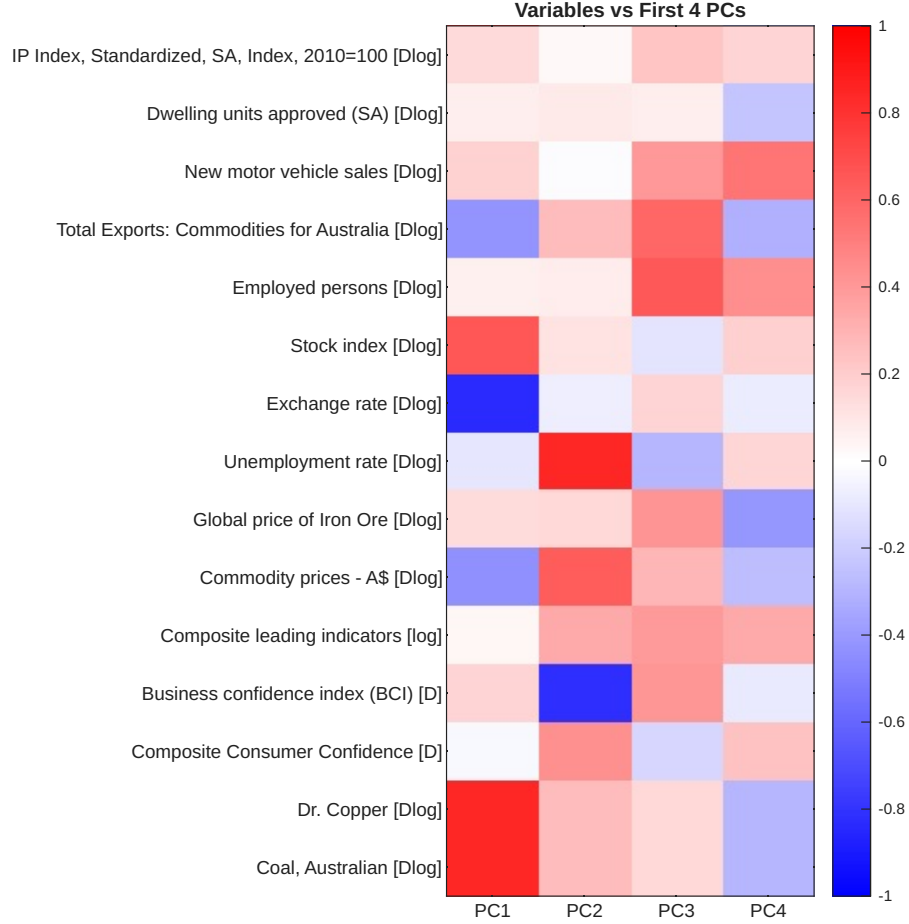


Figure 1: Four PCA Correlation Heatmap

The estimation pictured in the appendix supports our heuristic arguments, as the dominant loadings on the Principal Component 1, i.e. coal and copper prices, the exchange rate, and the ASX stock index, are crucial to the resource-exporting, terms-of-trade driven Australian economy. At the same time, these results also informed which of the strongly loading series we chose not to include in our final indicator set. Copper, for example, is a global price benchmark with essentially no Australia-specific channel, since Australia is not a major copper producer or trader, so its strong loading on f_t likely reflects its role as a generic barometer of the global commodity cycle rather than anything inherent to the Australian economy specifically. Coal and iron ore, in contrast, are Australia's two largest commodity exports by value, so the channel they represent is already captured by the exchange rate and by Australia's own export and commodity price series, without requiring a foreign benchmark price as a separate input. We therefore treat the high loadings of these series as confirmation that our chosen sectors are well represented and informative, while using that same evidence as a reason to exclude near duplicate or non-Australia-specific series, such as copper, rather than to add them.

3 Model Specification

Each (de-measured, differenced) series $y_{i,t}$ is assumed to load on a single common factor and an idiosyncratic AR(2) component,

$$y_{i,t} = \lambda_i f_t + e_{i,t}, \quad f_t = \phi_1 f_{t-1} + \phi_2 f_{t-2} + w_t, \quad e_{i,t} = \psi_{i,1} e_{i,t-1} + \psi_{i,2} e_{i,t-2} + \varepsilon_{i,t},$$

written in state-space form as $\mathbf{y}_t = \mathbf{H}\mathbf{h}_t + \mathbf{w}_t$, $\mathbf{h}_t = \mathbf{F}\mathbf{h}_{t-1} + \mathbf{v}_t$, $\mathbf{v}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{Q})$, with $\mathbf{h}_t$ the 27×1 state vector and $\mathbf{y}_t$ the 8×1 observation vector:

$$\mathbf{y}_t = \begin{pmatrix} \Delta \text{GDP}_t \\ \Delta \text{IP}_t \\ \Delta \text{Con}_t \\ \Delta \text{Veh}_t \\ \Delta \text{Emp}_t \\ \Delta \text{Stk}_t \\ \Delta \text{Exp}_t \\ \Delta \text{CLI}_t \end{pmatrix}, \quad \mathbf{h}_t = \begin{pmatrix} f_{t+3}, f_{t+2}, f_{t+1}, f_t, f_{t-1}, f_{t-2}, f_{t-3}, f_{t-4}, \\ e_{\text{GDP},t}, e_{\text{GDP},t-1}, e_{\text{GDP},t-2}, e_{\text{GDP},t-3}, e_{\text{GDP},t-4}, \\ e_{\text{IP},t}, e_{\text{IP},t-1}, e_{\text{Con},t}, e_{\text{Con},t-1}, e_{\text{Veh},t}, e_{\text{Veh},t-1}, \\ e_{\text{Emp},t}, e_{\text{Emp},t-1}, e_{\text{Stk},t}, e_{\text{Stk},t-1}, e_{\text{Exp},t}, e_{\text{Exp},t-1}, \\ e_{\text{CLI},t}, e_{\text{CLI},t-1} \end{pmatrix}_{27 \times 1}^\top.$$

The factor block is extended three steps into the future ($f_{t+3}, f_{t+2}, f_{t+1}$) purely so that the CLI can load on f_{t+3} as a named coordinate of today's state, while GDP – being a quarterly flow – continues to load on a five-month, triangularly-weighted window of monthly factor values ($\frac{1}{3}, \frac{2}{3}, 1, \frac{2}{3}, \frac{1}{3}$ on $f_t, \dots, f_{t-4}$), with the same weights applied to its own idiosyncratic component, following the same temporal-aggregation device used in the baseline code, in the spirit of Camacho and Pérez-Quiros's (2010) Euro-STING methodology. Every other indicator loads contemporaneously on f_t alone and carries a simple two-lag idiosyncratic AR(2). The explicit measurement and transition equations implementing this structure are given in Appendix A.

$$\mathbf{Z} = (\lambda_{\text{GDP}}, \lambda_{\text{IP}}, \lambda_{\text{Con}}, \lambda_{\text{Veh}}, \lambda_{\text{Emp}}, \lambda_{\text{Stk}}, \lambda_{\text{Exp}}, \lambda_{\text{CLI}}, \phi_1, \phi_2, \{\psi_{i,1}, \psi_{i,2}\}_{i \in \{\text{GDP}, \text{IP}, \text{Con}, \text{Veh}, \text{Emp}, \text{Stk}, \text{Exp}, \text{CLI}\}}, \{\sigma_{\varepsilon,i}\}_{i \in \{\text{GDP}, \dots, \text{CLI}\}})^\top$$

The full vector of freely estimated parameters has 34 parameters in total. Identification of the factor's scale is achieved by *fixing* the variance of the factor's own innovation at unity ($\sigma_w^2 \equiv 1$, not estimated) rather than by normalising any loading – λ_{GDP} is therefore freely estimated like every other loading, and is reported among the results below. The complete table of 34 point estimates and standard errors is given in Appendix C.

4 Results

`fminunc` converged to a local minimum after computing a finite-difference approximation to the Hessian for the Cramér–Rao standard errors. As an internal consistency check, the correlation between the full fitted GDP series (factor *and* idiosyncratic component) and observed GDP growth is, unsurprisingly, 1.000 at the months where GDP is actually observed, which confirms the filter reconstructs the data it was given. The correlation between observed GDP growth and the *factor-only* fitted series (idiosyncratic component switched off) is 0.9617, suggesting the common cycle alone explains the bulk of the co-movement in Australian GDP growth.

Table 2: Factor loading significance tests.

Variable	Factor loading	Std. error	t-stat	p-value
GDP	0.5995	0.0457	13.125	< 0.001
IP	0.0081	0.0439	0.184	0.854
Construction	0.1469	0.0442	3.324	0.001
Vehicle sales	0.3639	0.0458	7.937	< 0.001
Employment	0.8531	0.0652	13.084	< 0.001
Stock index	0.0155	0.0698	0.222	0.824
Exports	0.1396	0.0624	2.238	0.025
CLI	0.0087	0.0091	0.957	0.339

Employment and GDP itself carry by far the largest, most precisely estimated loadings, followed by vehicle sales and construction, all significant at conventional levels. Industrial production, the stock index and the CLI, by contrast, load only weakly on the common factor and are not statistically distinguishable from zero. This is a useful result in its own right, since it suggests that for this particular sample these three series move largely on their own idiosyncratic dynamics rather than tracking the broad Australian cycle, and would be reasonable candidates to reconsider in a refinement of the indicator set. As an internal consistency check, the correlation between the full fitted GDP series (factor *and* idiosyncratic component) and observed GDP growth is, unsurprisingly, 1.000 at the months where GDP is actually observed, which simply confirms the filter reconstructs the data it was given. More informatively, the correlation between observed GDP growth and the

factor-only fitted series (idiosyncratic component switched off) is 0.9617, suggesting the common cycle alone explains the bulk of the co-movement in Australian GDP growth.

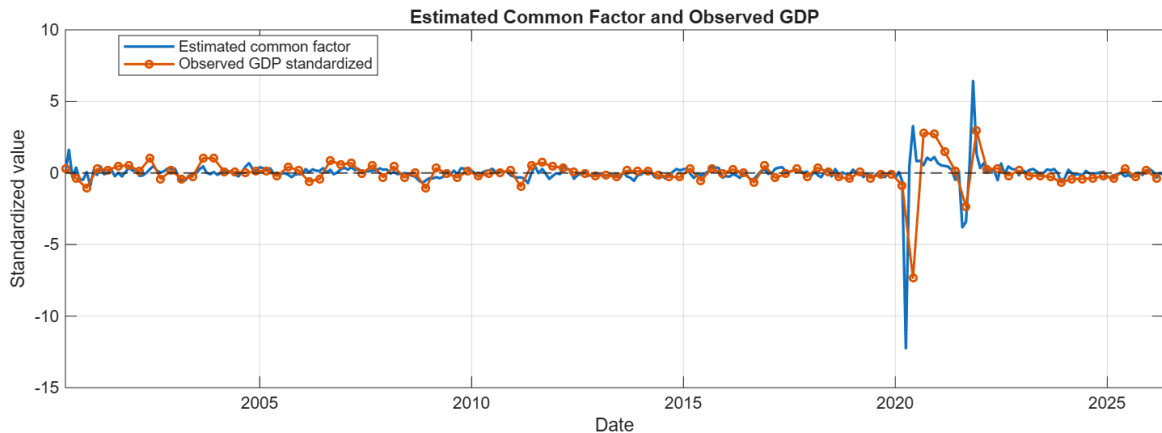


Figure 2: Estimated common factor versus standardized observed GDP growth.

Figure 2 compares the estimated latent factor with the standardized GDP growth. The two series track each other closely around the major turning points in our sample – most visibly the 2020 pandemic shock, which is reassuring: it suggests the factor is picking up genuine, broad-based cyclical movement rather than an artefact of any single series. Some divergence nonetheless remains in calmer periods.

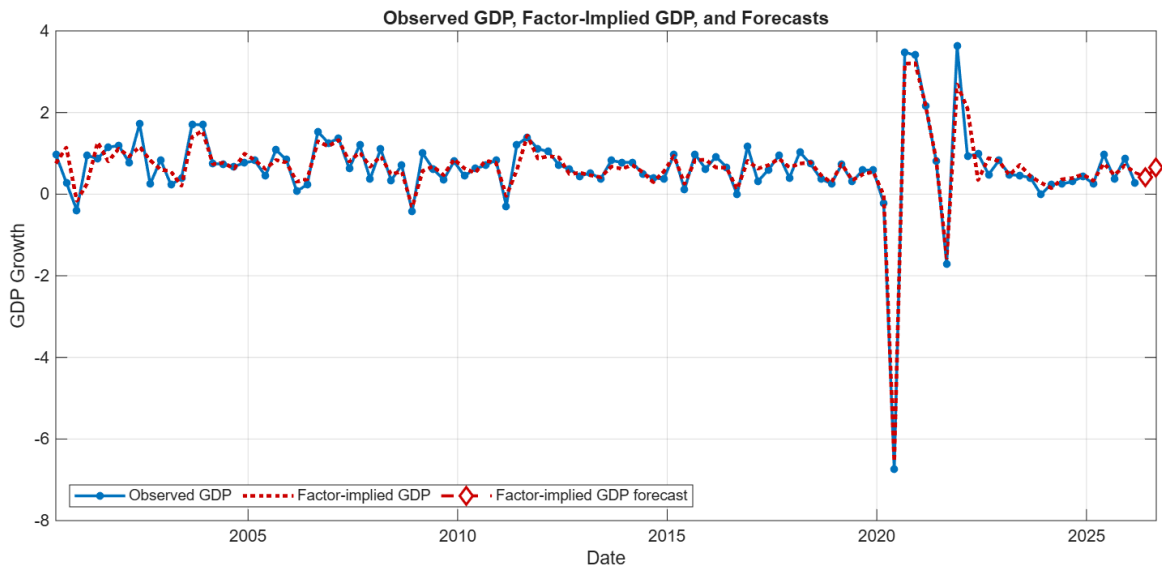


Figure 3: Observed GDP growth, factor-implied GDP growth and factor-implied forecasts.

Figure 3 goes a step further and compares the factor-only fitted series directly against *actually observed* quarterly GDP growth and the forecasts for the next two quarterly GDP releases. The co-movement is again strongest around recessions and sharp slowdowns, reinforcing the reading that the common factor captures much of the cyclical signal in Australian activity, even though it was extracted purely from the comovement of seven monthly indicators. The remaining gaps between the two lines likely reflect GDP-specific idiosyncratic noise not shared with the rest of the economy. Finally, the factor-only state forecast, as described in Section 3, implies Australian real GDP growth of **0.41%** for Q2-2026 and **0.65%** for Q3-2026.

5 Concluding remark

Overall, adapting the baseline Stock–Watson model to Australia required no change to the underlying estimation machinery, only to the content of the measurement and transition matrices, reflecting a different country, a richer indicator set, and one genuinely leading variable. The resulting factor appears to track the Australian cycle sensibly, particularly around major shocks, which we take as informal but encouraging evidence that the model is doing real economic work rather than simply curve-fitting.

B Figures

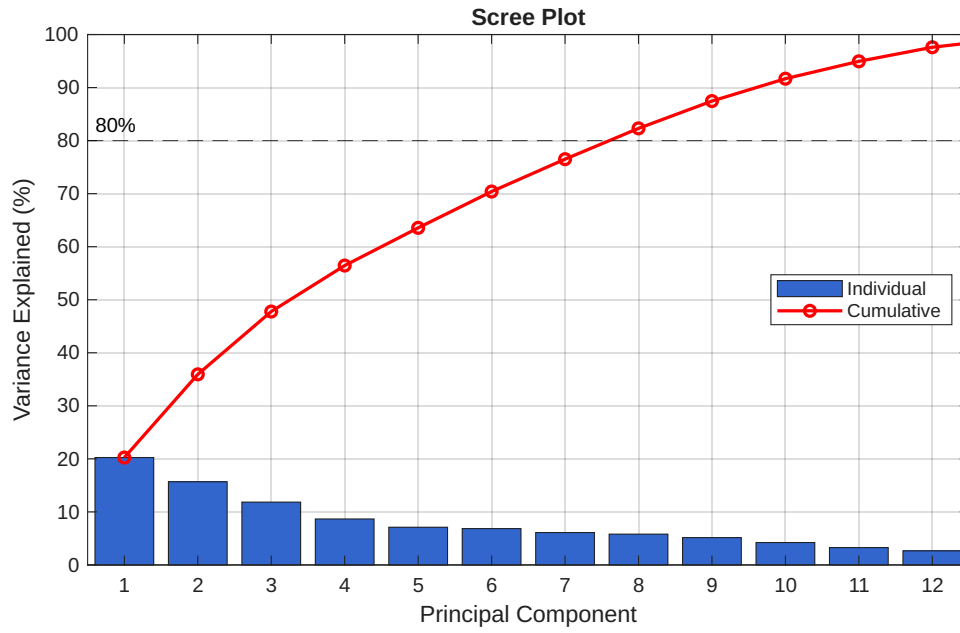


Figure 4: Principal Component Analysis Variance Explanation

C Full parameter estimates

Table 3: Maximum-likelihood estimates and Cramér–Rao standard errors for all 34 freely estimated parameters in $\mathbf{Z}$.

Param.	Estimate	Std. error	Param.	Estimate	Std. error
λ_{GDP}	0.5995	0.0457	$\psi_{\text{Veh},2}$	-0.4559	0.0544
λ_{IP}	0.0081	0.0439	$\psi_{\text{Emp},1}$	0.0548	0.1002
λ_{Con}	0.1469	0.0442	$\psi_{\text{Emp},2}$	0.3903	0.1357
λ_{Veh}	0.3639	0.0458	$\psi_{\text{Stk},1}$	0.1660	0.0617
λ_{Emp}	0.8531	0.0652	$\psi_{\text{Stk},2}$	-0.1046	0.0581
λ_{Stk}	0.0155	0.0698	$\psi_{\text{Exp},1}$	-0.0294	0.0589
λ_{Exp}	0.1396	0.0624	$\psi_{\text{Exp},2}$	-0.0849	0.0587
λ_{CLI}	0.0087	0.0091	$\psi_{\text{CLI},1}$	1.6694	0.0389
ϕ_1	0.1752	0.0904	$\psi_{\text{CLI},2}$	-0.7504	0.0388
ϕ_2	-0.2227	0.0853	σ_{GDP}	0.1590	0.0560
$\psi_{\text{GDP},1}$	0.6231	0.0935	σ_{IP}	0.6934	0.0288
$\psi_{\text{GDP},2}$	-0.8357	0.1013	σ_{Con}	0.8727	0.0364
$\psi_{\text{IP},1}$	0.8371	0.0600	σ_{Veh}	0.6494	0.0295
$\psi_{\text{IP},2}$	-0.1564	0.0605	σ_{Emp}	0.4910	0.0764
$\psi_{\text{Con},1}$	-0.5423	0.0574	σ_{Stk}	0.9791	0.0405
$\psi_{\text{Con},2}$	-0.2329	0.0575	σ_{Exp}	0.9979	0.0415
$\psi_{\text{Veh},1}$	-0.0533	0.0580	σ_{CLI}	0.1963	0.0082

Table 4: Correlation of Each Variable with the Estimated Common Factor (PC1)

Variable	Corr. with PC1
Coal, Australian [$\Delta \log$]	+0.845
Dr. Copper [$\Delta \log$]	+0.845
Exchange rate [$\Delta \log$]	-0.832
Stock index [$\Delta \log$]	+0.663
Commodity prices – A\$ [$\Delta \log$]	-0.434
Total Exports: Commodities for Australia [$\Delta \log$]	-0.414
New motor vehicle sales [$\Delta \log$]	+0.178
Business confidence index (BCI) [Δ]	+0.168
IP Index, Standardized, SA [$\Delta \log$]	+0.143
Global price of Iron Ore [$\Delta \log$]	+0.135
Unemployment rate [$\Delta \log$]	-0.095
Dwelling units approved (SA) [$\Delta \log$]	+0.063
Employed persons [$\Delta \log$]	+0.062
Composite leading indicators [$\log$]	+0.035
Composite Consumer Confidence [Δ]	-0.025